

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem
Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2640–2659; original printed p. 89; physical source-PDF pp. 78–79; recomposed running pp. 70–71. The compiled reader is the output of the same corrected French edition, not an independent original typescript witness. Line 2660 is blank; Corollary 2.2 begins at line 2661 and is excluded.

2 The Finiteness Theorem

Theorem 2.1⁽¹⁾. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X , and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. Suppose that X is locally embeddable in a regular prescheme^{*}. Let $i \in \mathbf{Z}$. Suppose that:*

a) *for every $x \in U = X - Y$, one has*

$$H^{i-c(x)}(\mathcal{F}_x) = 0,$$

where we have set⁽²⁾:

$$c(x) = \text{codim}(\overline{\{x\}} \cap Y, \overline{\{x\}}). \quad (2.1)$$

Then:

b) $\mathcal{H}_Y^i(\mathcal{F})$ *is coherent.*

⁽¹⁾*Editor's note.* For an analogous statement, but in a somewhat more general situation, see Mme Raynaud (Raynaud M., “Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental,” *Ann. Sci. Éc. Norm. Sup.* (4) **7** (1974), pp. 29–52, Proposition II.2.1).

^{*}This condition may be generalized to the hypothesis that a “dualizing complex,” in the sense defined in R. Hartshorne, *Residues and Duality* (cited in the note to Exposé IV, p. 46), exists locally on X .

⁽²⁾*Editor's note.* As in Exposé V, $H^*(\mathcal{F}_x)$ denotes the local cohomology $H_{\mathfrak{m}_x}^*(\mathcal{F}_x)$.